

9	(a) time for number of atoms / nuclei / activity (of the isotope) to be reduced to one half (of its initial value)	M1	A1	[2]
(b)	(i) $A = \lambda N$ $460 = N \times \ln 2 / (8.1 \times 24 \times 60 \times 60)$ $N = 4.6 \times 10^8$	C1	C1	A1
				[3]
	(ii) number of water molecules in 1.0 kg $= (6.02 \times 10^{23}) / (18 \times 10^{-3})$ $= 3.3 \times 10^{25}$	C1		
	ratio $= (3.3 \times 10^{25}) / (4.6 \times 10^8)$ $= 7.2 (7.3) \times 10^{16}$		A1	[2]

Page 5	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – May/June 2014	9702	42

- (c) $A = A_0 e^{-\lambda t}$ and $\lambda t_{1/2} = \ln 2$ C1
- $170 = 460 \exp(-\{\ln 2\} t / 8.1)$ C1
- $t = 11.6$ days (*allow 2 s.f.*) A1 [3]

Section B